

CHANY, Russian Federation

WMO: 296020

Lat: 55.3168N

Lon: 76.7445E

Elev: 365

StdP: 14.50

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.2	-27.3	-38.6	0.6	-33.0	-32.9	0.9	-27.0	20.5	14.7	18.3	14.2	3.9	340	0.646

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	85.3	65.8	82.3	65.1	79.5	63.7	69.0	79.4	67.3	77.8	65.7	75.9	8.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.5	95.3	73.3	63.6	89.2	71.7	61.9	83.9	70.3	33.5	79.2	32.1	78.0	30.8	76.0	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.8	18.2	16.1	DB	-36.2	90.2	8.3	3.5	-42.2	92.7	-47.0	94.8	-51.7	96.8	-57.7	99.4	(4)
(5)			WB	-36.4	71.7	7.9	2.2	-42.1	73.3	-46.7	74.6	-51.1	75.8	-56.8	77.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.9	-1.9	2.9	19.9	39.5	53.3	63.1	66.1	61.5	50.9	37.5	19.0	5.2		
	DBStd	26.23	14.85	13.43	10.63	10.58	9.86	7.88	6.20	6.91	8.44	8.90	14.02	14.73		
	HDD50	7104	1610	1320	932	344	82	7	0	3	89	399	929	1389		
	HDD65	11216	2075	1740	1397	766	379	126	64	152	430	854	1379	1854		
	CDD50	1597	0	0	0	28	185	399	500	359	116	10	0	0		
	CDD65	234	0	0	0	0	17	68	99	43	7	0	0	0		
	CDH74	2249	0	0	0	5	243	653	850	421	77	0	0	0		
	CDH80	533	0	0	0	0	53	167	206	93	14	0	0	0		
(14) Wind	WSAvg	7.6	6.8	7.2	8.2	8.7	8.7	7.2	6.8	6.9	7.4	7.7	8.1	7.6		
(15) Precipitation	PrecAvg	15.0	0.7	0.6	0.6	0.7	1.0	2.1	2.6	2.0	1.3	1.0	1.3	1.0		
	PrecMax	21.8	1.5	1.1	1.5	1.7	2.0	4.8	5.9	3.9	2.9	2.4	3.0	1.8		
	PrecMin	10.7	0.1	0.2	0.0	0.1	0.4	0.6	0.7	0.4	0.4	0.3	0.2	0.4		
	PrecStd	3.1	0.4	0.2	0.4	0.4	0.6	1.0	1.3	0.8	0.7	0.6	0.7	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.5	33.6	43.5	73.1	86.0	88.2	89.3	87.0	81.4	65.9	44.7	34.4		
		MCWB	31.8	31.9	37.9	55.3	61.6	65.9	68.6	66.2	60.5	51.9	41.6	32.9		
	2%	DB	28.3	30.4	39.3	66.1	80.4	84.7	85.0	82.5	75.4	60.5	41.7	32.2		
		MCWB	27.3	29.0	35.4	52.4	60.6	64.7	67.3	65.5	59.0	49.3	39.4	31.3		
	5%	DB	24.5	26.8	36.5	61.4	76.1	81.3	82.0	78.8	70.0	55.1	38.2	28.7		
		MCWB	23.6	25.7	33.7	50.3	58.2	64.1	66.3	63.1	56.8	46.7	36.2	27.7		
	10%	DB	20.3	22.7	34.3	55.8	71.0	77.6	79.0	75.1	65.0	50.4	34.6	24.9		
		MCWB	19.4	21.6	32.0	46.6	56.0	62.9	65.5	62.5	54.2	44.4	32.8	23.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.9	32.3	38.7	57.2	65.2	70.1	71.8	69.4	63.2	53.6	42.8	33.6		
		MCDB	32.5	32.9	42.2	68.7	80.8	82.1	83.5	80.6	76.6	62.4	43.7	34.0		
	2%	WB	27.4	29.1	35.9	54.1	62.0	67.5	69.8	67.1	60.3	50.6	39.4	31.1		
		MCDB	28.2	30.3	38.5	64.4	77.4	78.9	79.6	78.3	71.8	58.8	41.6	32.2		
	5%	WB	23.7	25.8	34.1	50.5	59.5	65.8	68.3	65.3	57.8	47.5	36.2	27.8		
		MCDB	24.5	26.9	36.2	60.1	73.3	77.0	78.2	75.4	67.9	54.2	38.0	28.8		
	10%	WB	19.4	21.7	32.2	47.4	57.0	64.1	66.6	63.4	55.2	44.7	33.1	23.9		
		MCDB	20.1	22.7	34.3	55.3	69.0	74.6	76.3	72.5	64.0	50.1	34.2	24.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.9	15.2	15.4	16.4	20.4	18.9	18.4	18.9	17.7	13.3	11.0	13.5		
		MCDBR	14.4	13.2	13.8	24.3	27.1	24.3	23.4	25.2	25.1	21.5	11.0	14.3		
		MCWBR	14.1	12.8	11.0	13.8	11.6	9.4	8.9	10.5	11.6	12.5	9.2	14.1		
		MCDBR	14.1	13.1	12.3	22.9	24.8	21.1	20.2	21.1	22.8	19.3	10.5	14.5		
	5% WB	MCWBR	13.9	12.9	10.3	13.7	12.5	9.4	9.0	9.9	11.5	12.0	9.3	14.3		
(40) Clear-Sky Solar Irradiance	taub		0.284	0.316	0.374	0.406	0.401	0.409	0.431	0.419	0.365	0.352	0.320	0.287		
	taud		2.033	2.004	1.926	2.112	2.233	2.286	2.226	2.228	2.340	2.229	2.079	1.983		
	Ebn at Noon		190	229	242	256	267	266	256	251	251	218	172	153		
	Edh at Noon		22	32	45	43	41	39	41	38	30	26	21	18		
(44) All-Sky Solar Radiation	RadAvg		301	645	1149	1470	1735	1879	1721	1391	983	487	235	195		
(45)	RadStd		23	32	63	125	154	162	116	133	96	58	39	21		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page